PHYSICS

1. What quantity remains same in ideal transformer.
(1) voltage (2) Current
(3) Power (4) None of these

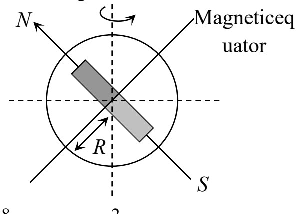
2. An electrons moving in a circular orbit of radius r makes υ rotations per second. The magnetic moment of the orbital electron is
(1) Zero (2) $\pi r^2 \upsilon e$
(3) $\pi r^2 \upsilon^2 e$ (4) $\frac{r^2 \upsilon e}{2\pi}$

3. At a place the earth's horizontal component of magnetic field is 0.38×10^{-4} weber / m^2. If the angle of dip at that place is 60^0, then the vertical component of earth's field at that place in weber/m^2 will be approximately
(1) 0.12×10^{-4} (2) 0.24×10^{-4}
(3) 0.40×10^{-4} (4) 0.62×10^{-4}

4. A dip circle is so set that it moves freely in the magnetic meridian. In this position, the angle of dip is 40^0. Now the dip circle is rotated so that the plane in which the needle moves makes an angle of 30^0 with the magnetic meridian. In this position, the needle will dip by the angle
(1) 40^0 (2) 30^0
(3) More than 40^0 (4) Less than 40^0

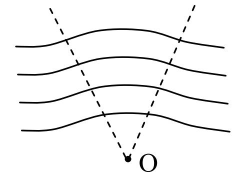
5. A steal wire of length *l* has a magnetic moment M. It is bent in L shape at its midpoint. The new magnetic moment is
(1) M (2) $\frac{M}{\sqrt{2}}$
(3) $\frac{M}{2}$ (4) 2M

6. Earth's magnetic field may be supposed to be due to a small bar magnet located at the centre of the earth. If the magnetic field at a point on the magnetic equator is 0.3×10^{-4} T. Magnet moment of bar magnet is



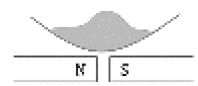
(1) 7.8×10^8 amp $\times$ m^2
(2) 7.8×10^{22} amp $\times$ m^2
(3) 6.4×10^{22} amp $\times$ m^2
(4) 6.4×10^8 amp$\times$m^2

7. If the magnetic lines of force are shaped like arcs of concentric circles with their centre at point O in a certain section of a magnetic field:



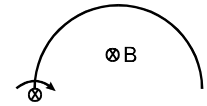
(1) The intensity of the field in this section should at each point be inversely proportional to its distance from point O
(2) The intensity of the field in this section should at each point be inversely proportional to square of its distance from point O
(3) The intensity of the field in this section should at each point be inversely proportional to cube of its distance from point O
(4) Nothing can be said

8. The liquid in the watch glass in the following figure is –



(1) Ferromagnetic
(2) Paramagnetic
(3) Diamagnetic
(4) Nonmagnetic

9. A semicircular wire of radius R is rotated with constant angular velocity ω about an axis passing through one end and perpendicular to the plane of the wire. There is a uniform magnetic field of strength B. The induced e.m.f. between the ends is:



(1) B $\omega R^2/2$ (2) 2 B ωR^2
(3) is variable (4) none of these

10. A paramagnetic sample shows a net magnetization of 6 A/m when it is placed in an external magnetic field of 0.4 T at a temperature of 4 K. When the sample is placed in an external magnetic field of 0.3 T at a temperature of 24 K, then the magnetization will be
(1) 4 A/m (2) 0.75 A/m
(3) 2.25 A/m (4) 1 A/m

11. An electron moving around the nucleus with an angular momentum l has a magnetic moment

(1) $\frac{e}{m}L$ (2) $\frac{e}{2m}L$

(3) $\frac{2e}{m}L$ (4) $\frac{e}{2\pi m}L$

12. Choose the correct statement

(1) A paramagnetic material tends to move from a strong magnetic field to weak magnetic filed.

(2) A magnetic material is in the paramagnetic phase below its curie temperature.

(3) The resultant magnetic moment in an atom of a diamagnetic substance is zero

(4) Typical domain size of a ferromagnetic material is 1 nm.

13. The electric and magnetic field of an electromagnetic wave are :

(1) in phase and parallel to each other

(2) in opposite phase and perpendicular to each other

(3) in opposite phase and parallel to each other

(4) in phase and perpendicular to each other.

14. A plane electromagnetic wave propagating in the X-direction has wavelength of 6.0 mm. the electric field is in the Y-direction and its maximum magnitude is 33 Vm^{-1} . The equation for the electric field as a function of x and t is:

(1) $11\sin\pi\left(t-\frac{x}{c}\right)$

(2) $33\sin\pi \times 10^{11}\left(t-\frac{x}{c}\right)$

(3) $33\sin\pi\left(t-\frac{x}{c}\right)$

(4) $11\sin\pi \times 10^{11}\left(t-\frac{x}{c}\right)$

15. The relative permeability (μ_r) of a substance is related to its susceptibility (x) as

(1) $\mu_r = 1 - \chi$ (2) $\mu_r = 1 + \chi$

(3) $\mu_r = 1 - \chi^2$ (4) $\mu_r = 1 + \chi^2$

16. If N is the number of turns in a coil, the value of self-inductance varies as

(1) N^0 (2) N

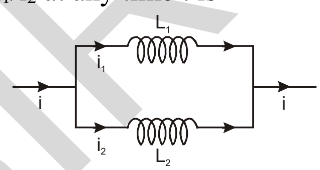
(3) N^2 (4) N^{-2}

17. The mean electric energy density between plates of a charged capacitor:

(1) $\frac{q^2}{2\varepsilon_0 A^2}$ (2) $\frac{q}{2\varepsilon_0 A^2}$

(3) $\frac{q^2}{2\varepsilon_0 A}$ (4) None of these

18. Two inductors L_1 and L_2 are connected in parallel and a time varying current i flows as shown. The ratio of currents i_1/i_2 at any time t is



(1) L_1/L_2 (2) L_2/L_1

(3) $\frac{L_1^2}{(L_1+L_2)^2}$ (4) $\frac{L_2^2}{(L_1+L_2)^2}$

19. Hysteresis is shown by which of the following material

(1) Diamagnetic (2) Ferromagnetic

(3) Paramagnetic (4) All of these

20. If the susceptibility of dia. para and ferro magnetic materials are x_d, x_P, x_f respectively, then.

(1) $x_d < x_p < x_f$ (2) $x_d < x_f < x_p$

(3) $x_f < x_d < x_p$ (4) $x_f < x_p < x_d$

21. A paramagnetic liquid is taken in a U-tube and arranged so that one of its limbs is kept between pole pieces of the magnet, the liquid level in the limb.

(1) Goes down

(2) Rises up

(3) Remain same

(4) First goes down and then rises

22. If the wavelength of light is 4000Å, the number of waves in 1 mm length will be:

(1) 25 (2) 0.25

(3) 0.25×10^4 (4) 2.5×10^4

23. The net magnetic flux through any closed surface, kept in a magnetic field is

(1) Zero (2) $\frac{\mu_0}{4\pi}$

(3) $4\pi\mu_0$ (4) $\frac{4\mu_0}{\pi}$

24. A coil having an inductance of 0.5 G carries a current which is uniformly varying from zero to 10 ampere in 2 second. The emf (in volt) generated in the coil is

(1) 10 (2) 5

(3) 2.5 (4) 1.25

25. A plane electromagnetic wave is incident on material surface. The wave delivers momentum 'P' and energy 'E'

(1) $P \neq 0, E \neq 0$ (2) $P = 0, E = 0$

(3) $P = 0, E \neq 0$ (4) $P \neq 0, E = 0$

26. In an ac circuit, the reactance equal to the resistance. The power factor of the circuit will be

(1) 1 (2) $\frac{1}{2}$

(3) $\frac{1}{\sqrt{2}}$ (4) Zero

27. An alternating supply of 220 V is applied across a circuit with resistance 22 Ω and impedance 44 Ω. The power dissipated in the circuit will be -

(1) 1100 W (2) 550 W

(3) 2200 W (4) (2200/3) W

28. The power factor of LCR circuit at resonance is

(1) 0 (2) $\frac{1}{2}$

(3) $\frac{1}{\sqrt{2}}$ (4) 1

29. The core of a transformer is usually made of

(1) Steel (2) Copper

(3) Soft iron (4) Aluminium

30. Needles N_1, N_2 and N_3 are made of a ferromagnetic, a paramagnetic and a diamagnetic substance respectively. A magnet when brought close to them will.

(1) Attract N_1 and N_2 strongly but repel N_3

(2) Attract N_1 strongly, N_2 weakly and repel weakly N_3

(3) Attract N_1 strongly, but repel N_2 and N_3 weakly

(4) Attract all three of them

31. Find the energy stored in a 60 cm length of a laser beam operating at 4 mW.

(1) 8×10^{-12}J (2) 6×10^{-12}J

(3) 4×10^{-12}J (4) 7×10^{-12}J

32. A bar magnet is demagnetized by inserting it inside a solenoid of length 0.2 m, 100 turns, and carrying a current of 5.2 A. The coercivity of the bar magnet is ;

(1) 1200 A/m

(2) 2600 A/m

(3) 520 A/m

(4) 285 A/m

33. Match list I (Electromagnetic wave type) with List II (Its association / application) and select the correct option from the choices given below the lists.

List I	List II
(p) Infrared wave	(i) To treat muscular strain
(q) Radio waves	(ii) For broadcasting
(r) X – rays	(iii) To detect fracture for bones
(s) Ultraviolet rays	(iv) Absorbed by the ozone layer of the atmosphere

Code

	(p)	**(q)**	**(r)**	**(s)**
(1)	(iii)	(ii)	(i)	(iv)
(2)	(i)	(ii)	(iii)	(iv)
(3)	(iv)	(iii)	(ii)	(i)
(4)	(i)	(ii)	(iv)	(iii)

34. Lorentz force can be calculated by using the formula

(1) $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

(2) $\vec{F} = q(\vec{E} - \vec{v} \times \vec{B})$

(3) $\vec{F} = q(\vec{E} + \vec{v} . \vec{B})$

(4) $\vec{F} = q(\vec{E} \times \vec{v} \times \vec{B})$

35. In a plane electromagnetic wave, the electric field oscillates sinusoidally at a frequency of 2.5×10^{10} Hz and amplitude 480 V/m. The amplitude of the oscillating magnetic field will be
(1) 1.52×10^{-8} Wb/m^2
(2) 1.52×10^{-7} Wb/m^2
(3) 1.6×10^{-6} Wb/m^2
(4) 1.6×10^{-7} Wb/m^2

36. If ε_0 and μ_0 represent the permittivity and permeability of vacuum and ε and μ represent the permittivity and permeability of medium, the refractive index of the medium is given by
(1) $\sqrt{\frac{\varepsilon_0\mu_0}{\varepsilon\mu}}$ (2) $\sqrt{\frac{\varepsilon\mu}{\varepsilon_0\mu_0}}$
(3) $\sqrt{\frac{\varepsilon}{\mu_0\varepsilon_0}}$ (4) $\sqrt{\frac{\mu_0\varepsilon_0}{\varepsilon}}$

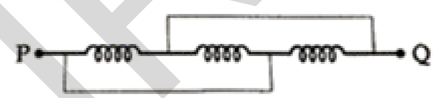
37. An infinitely long cylinder is kept parallel to a uniform magnetic field B directed along positive z – axis. The direction of induced current as seen from the z axis will be:
(1) Clockwise of the +ve z-axis
(2) Anticlockwise of the +ve z – axis
(3) Zero
(4) Along the magnetic field

38. Arrange the following electromagnetic radiations in the order of increasing energy:
A : Blue light B : Yellow light
C : X – rays D : Radio wave
(1) A, B, D, C
(2) C, A, B, D
(3) B, A, D, C
(4) D, B, A, C

39. When 100 volt dc is applied across a solenoid, a current of 1.0 amp flows in it. When 100 volt ac is applied across the same coil, the current drops to 0.5 amp. If the frequency of ac source is 50 Hz the impedance and inductance of the solenoid are
(1) 200 ohms and 0.55 henry
(2) 100 ohms and 0.86 henry
(3) 200 ohms and 1.0 henry
(4) 100 ohms and 0.93 henry

40. A sinusoidal voltage $V_o \sin\omega t$ is applied across a series combination of resistance R and inductor L. The amplitude of the current in the circuit is
(1) $\frac{V_o}{\sqrt{R^2+\omega^2L^2}}$
(2) $\frac{V_o}{\sqrt{R^2-\omega^2L^2}}$
(3) $\frac{V_o}{\sqrt{R^2+\omega^2L^2}}\sin\omega t$
(4) V_o/R

41. Pure inductance of 3.0 H is connected as shown below. The equivalent inductance of the circuit is



(1) 1 H (2) 2 H
(3) 3 H (4) 9 H

42. Which of the following is not an application of eddy currents
(1) Induction furnace
(2) Galvanometer damping
(3) Speedometer of automobiles
(4) X -ray crystallography

43. Quantity that remains unchanged in a transformer is
(1) Voltage (2) Current
(3) Frequency (4) None of above

44. In general the wavelength of microwaves is:
(1) More than that of infrared waves
(2) More than that of radio waves
(3) Less than that of infrared waves
(4) Less than that of ultraviolet waves

45. Which of the following waves have the maximum wavelength?
(1) X – rays (2) I.R. rays
(3) UV rays (4) Radio waves

CHEMISTRY

46. What is the oxidation state of Fe in $[Fe(H_2O)_5(NO)]^{2+}$ ion?
(1) +2 (2) +3
(3) +1 (4) 0

47. Which gives only 25% mole of AgCl, when reacts with $AgNO_3$:–
(1) $PtCl_2.4NH_3$ (2) $PtCl_4.5NH_3$
(3) $PtCl_4.4NH_3$ (4) $PtCl_4.3NH_3$

48. A reagent used for identified nickel ion is:–
(1) Potassium ferrocyanide
(2) Phenolphthalein
(3) Dimethyl glyoxime
(4) EDTA

49. A planar complex (Mabcd) gives:–
(1) Two optical isomer
(2) Two geometrical isomers
(3) Three optical isomers
(4) Three geometrical isomers

50. What is incorrect for $K_4[Fe(CN)_6]$
(1) Oxidation number of Iron is +2
(2) It exhibits diamagnetic character
(3) It exhibits paramagnetic character
(4) It involves d^2sp^3 hybridisation

51. Which of the following complex does not show geometrical isomerism?
(1) $[CO(NH_3)_4Cl_2]^+$
(2) $[CO(NH_3)_3(NO_2)_3]$
(3) $[Cr(en)_3]^{3+}$
(4) $[Pt(gly)_2]$

52. Which of the following has least conductivity in aqueous solution.
(1) $CoCl_3.4NH_3$ (2) $CoCl_3.3NH_3$
(3) $CoCl_3.5NH_3$ (4) $CoCl_3.6NH_3$

53. Which one is a heteroleptic complex?
(1) Ferrocene (2) Chromocene
(3) Prussian blue (4) Zeise's salt

54. The product of oxidation of I^- with MnO_4^- in weak alkaline medium is:
(1) IO_3^- (2) I_2
(3) IO^- (4) IO_4^-

55. The correct IUPAC name of complex, $[Rh(en)_2(ONO)(SCN)]NO_3$ is :
(1) diethane-1, 2-diamine nitrito-O-thiocyanato-S-rhodium (III) nitrate
(2) bis(ethane-1, 2-diamine) nitrito-O-thiocyanato-S-rhodium(III) nitrate
(3) bis(ethane-1, 2-diamine) nitrito-O-thiocyanato-S-rhodate(III) nitrate
(4) bis(ethane-1, 2-diamine) nitrito-N-thiocyanato-N-rhodium(II) nitrate.

56. EAN of the central metal in the complexes – $K_2[Ni(CN_4)]$, $[Cu(NH_3)_4]SO_4$ and $K_2[PtCl_6]$ are respectively:
(1) 36, 35, 86 (2) 34, 35, 84
(3) 34, 35, 86 (4) 34, 36, 86

57. All the following complex ions are found to be paramagnetic :
P : $[FeF_6]^{3-}$ Q : $[CoF_6]^{3-}$
R : $[V(H_2O)_6]^{3+}$ S : $[Ti(H_2O)_6]^{3+}$
(1) P > Q > R > S (2) P < Q < R < S
(3) P = Q = R = S (4) P > R > Q > S

58. Which of the following statements are wrong ?
(a) Al_4C_3 is an organometallic compounds
(b) Metal carbonyls are organometallic compounds
(c) TEL is π-bonded organometallic compound
(d) Frankland reagent is π-bonded organometallic compound
(1) a, b and d (2) a and c
(3) b and c (4) a, b, c and d

59. Atomic number of Cr and Fe are respectively 24 and 26, which of the following is paramagnetic with the spin of electron?
(1) $[Cr(CO)_6]$ (2) $[Fe(CO)_5]$
(3) $[Fe(CN)_6]^{4-}$ (4) $[Cr(NH_3)_6]^{3+}$

60. **Assertion:** In complex, $[Co(NH_3)_5(CO_3)]Cl$, the oxidation state of cobalt is +3.
Reason: Carbonate ligand is a monodentate bivalent anion.
(1) Both assertion and reason are correct, and the reason is the correct explanation for the assertion.
(2) Both assertion and reason are correct, but the reason is not the correct explanation for the assertion.
(3) The assertion is incorrect, but the reason is correct.
(4) Both are assertion and reason are incorrect.

61. Which of the following name formula combinations is not correct?

	Formula	Name
(1)	$K_2[Pt(CN)_4]$	Potasium tetracyanoplatinate (II)
(2)	$[Mn(CN)_5]^{2-}$	Pentacyanomagnate (II) ion
(3)	$K[Cr(NH_3)_2Cl_4]$	Potassium diammine tetrachloro chromate (III)
(4)	$[Co(NH_3)_4(H_2O)I]SO_4$	Tetraammine aquaiodo cobalt (III) sulphate

62. Which of the following has no unpaired electrons but is coloured?

(1) $K_2Cr_2O_7$ (2) K_2MnO_4

(3) $CuSO_4.5H_2O$ (4) $MnCl_2$

63. Which of the following oxidations state is common for all lanthanoids?

(1) +2 (2) +3

(3) +4 (4) +5

64. When acidified $K_2Cr_2O_7$ solution is added to Sn^{2+} salts, then Sn^{2+} changes to

(1) Sn (2) Sn^{3+}

(3) Sn^{4+} (4) Sn^{+}

65. Transition metals make the most efficient catalysts because of their ability to

(1) Adopt multiple oxidation states and to form complexes

(2) Form coloured ions

(3) Show paramagnetic due to unpaired electrons

(4) Form a large number of oxides.

66. Fe^{3+} ion is more stable than Fe^{2+} ion because

(1) More the charge on the atom, more is its stability

(2) Configuration of Fe^{2+} is $3d^6$ while Fe^{3+} is $3d^5$

(3) Fe^{2+} has a larger size than Fe^{3+}

(4) Fe^{3+} ions are coloured hence more stable.

67. Cu^{+} ion is not stable in aqueous solution because

(1) Second ionization enthalpy of copper is less than the first ionization enthalpy

(2) Large value of second ionization enthalpy of copper is compensated by much more negative hydration energy of $Cu^{2+}_{(aq)}$

(3) Hydration energy of $Cu^{+}_{(aq)}$ is much more negative than that of $Cu^{2+}_{(aq)}$

(4) Many copper (I) compounds are unstable in aqueous solution and undergo disproportionation reaction

68. What happens when potassium iodide reacts with acidic solution of potassium dichromate?

(1) It liberates iodine

(2) Potassium sulphate is formed

(3) Chromium sulphate is formed

(4) All the above products are formed

69. Which of the following are basic oxides?

$Mn_2O_7, V_2O_3, V_2O_5, CrO, Cr_2O_3$

(1) Mn_2O_7 and V_2O_3

(2) V_2O_3 and CrO

(3) CrO and Cr_2O_3

(4) V_2O_5 and V_2O_3

70. Match the column I with column II and mark the appropriate choice.

	Column I		Column II
(A)	$FeSO_4.7H_2O$	(i)	Green
(B)	$NiCl_2.4H_2O$	(ii)	Light pink
(C)	$MnCl_2.4H_2O$	(iii)	Pale green
(D)	$CoCl_2.6H_2O$	(iv)	Pink
(E)	Cu_2Cl_2	(v)	Colourless

(1) (A) $\to$ (iii), (B) $\to$ (iv), (C) $\to$ (i), (D) $\to$ (ii), (E) $\to$ (v)

(2) (A) $\to$ (ii), (B) $\to$ (iii), (C) $\to$ (iv), (D) $\to$ (i), (E) $\to$ (v)

(3) (A) $\to$ (v), (B) $\to$ (ii), (C) $\to$ (iii), (D) $\to$ (iv), (E) $\to$ (i)

(4) (A) $\to$ (iii), (B) $\to$ (i), (C) $\to$ (ii), (D) $\to$ (iv), (E) $\to$ (v)

71. Match the column I with column II and mark the appropriate choice.

Column I		Column II	
(A)	Green vitriol	(i)	$ZnSO_4.7H_2O$
(B)	Mohr's salt	(ii)	$FeSO_4.7H_2O$
(C)	Blue vitriol	(iii)	$FeSO_4.(NH_4)_2SO_4.6H_2O$
(D)	White vitriol	(iv)	$CuSO_4.5H_2O$

(1) (A)→(iii), (B)→(iv), (C)→(i), (D)→(ii)
(2) (A) → (ii), (B) → (iii), (C) → (iv), (D) → (i)
(3) (A) → (iv), (B) → (i), (C) → (ii), (D) → (iii)
(4) (A) → (i), (B) → (iv), (C) → (ii), (D) → (iii)

72. The melting point of Zn is lower as compared to those of the other elements of 3d series because :
(1) the d-orbitals are completely filled.
(2) the d-orbitals are partially filled.
(3) d-electrons do not participate in metallic bonding.
(4) (1) and (3) both.

73. Low oxidation states are found in transition elements when a complex compound has :
(1) ligands capable of π acceptor character.
(2) ligands capable of σ donor character.
(3) ligands capable of π acceptor character as well as σ donor character.
(4) ligands incapable of π acceptor character as well as σ donar character.

74. Which of the following statements is false ?
(1) An acidified solution of $K_2Cr_2O_7$ liberates iodine from potassium iodide
(2) In acidic solution, dichromate ions are converted to chromate ions.
(3) Potassium dichromate on heating undergoes decomposition to give Cr_2O_3 and O_2 gas.
(4) Potassium dichromate is used as a titrant of Fe^{2+} ion.

75. Ti^{2+} is purple while Ti^{4+} is colourless, because
(1) There is zero no crystal field effect in Ti^{4+}.
(2) Ti^{2+} has $3d^2$ configuration
(3) Ti^{4+} has $3d^2$ configuration
(4) Ti^{4+} is a very small cation when compared to Ti^{2+} and hence, does not absorb any radiation

76. Which one of the following compounds is expected to be coloured?
(1) $TiCl_3$ (2) Cu_2Cl_2
(3) $ZnSO_4$ (4) $ScCl_3$

77. Which of the following is not the configuration of lanthanoid?
(1) $[Xe]4f^{10}6s^2$
(2) $[Xe]4f^15d^16s^2$
(3) $[Xe]4f^{14}5d^{10}6s^1$
(4) $[Xe]4f^{14}5d^16s^2$

78. What is not true about ligand?
(1) It can act as Lew is base
(2) A monodentate ligand cannot cause chelation
(3) A multidentate ligand cannot cause chelation
(4) It can have no. of donor sites ≥ 1

79. What is the EAN of central metal in $[Ni(gly)_2]$ (At. No. of Ni = 28)
(1) 30 (2) 34
(3) 36 (4) 32

80. Which among the following complexes is diamagnetic?
(1) $[Cr(NH_3)_6]^{3+}$ (2) $[CoF_6]^{3-}$
(3) $Ni(CO)_4$ (4) $[CuCl_4]^{2-}$

81. Which of the following is not optically active?
(1) $[Co(en)_3]^{3+}$
(2) $[Cr(ox)_3]^{3-}$
(3) Cis-$[CoCl_2(en)_2]^+$
(4) Trans-$[CoCl_2(en)_2]^+$

82. Which one is an outer orbital complex?
(1) $[Ni(NH_3)_6]^{2+}$ (2) $[Mn(CN)_6]^{4-}$
(3) $[Co(NH_3)_6]^{3+}$ (4) $[Fe(CN)_6]^{4-}$

83. In the complex $[CoCl_2(en)_2]Br$, the co-ordination number and oxidation state of cobalt are :
(1) 6 and +3 (2) 3 and +3
(3) 4 and +2 (4) 6 and +1

84. Which of the following is a high spin complex?
(1) $[Co(NH_3)_6]^{3+}$ (2) $[Fe(CN)_6]^{4-}$
(3) $[Ni(CN)_4]^{2-}$ (4) $[FeF_6]^{3-}$

85. Which of the following factors tends to increase the stability of metal ion complexes?

(1) Higher ionic radius of the metal ion

(2) Higher charge/size ratio of the metal ion

(3) Lower ionisation potential of the metal ion

(4) Lower basicity of the ligand

86. Match Column-I with Column-II and select the correct answer with respect to hybridisation using the codes given below:

Column - I (Complex)		**Column - II (Hybridisation)**	
(I)	$[Au\ F_4]^-$	**(p)**	dsp^2 hybridisation
(II)	$[Cu(CN)_4]^{3-}$	**(q)**	sp^3 hybridisation
(III)	$[Co(C_2O_4)_3]^{3-}$	**(r)**	sp^3d^2 hybridisation
(IV)	$[Fe(H_2O)_5NO]^{2+}$	**(s)**	d^2sp^3 hybridisation

Codes :

	(I)	(II)	(III)	(IV)
(1)	q	p	r	s
(2)	p	q	s	r
(3)	p	q	r	s
(4)	q	p	s	r

87. The bond length in CO is 1.128 Å. What will be the bond length of CO in $Fe(CO)_5$?

(1) 1.158 Å (2) 1.128 Å

(3) 1.178 Å (4) 1.118 Å

88. {en = $H_2NCH_2CH_2NH_2$; atomic numbers : Ti = 22; Cr = 24; Cp = 27; Pt = 78}

	List-I		**List-II**
P.	$[Cr(NH_3)_4Cl_2)Cl]$	**1.**	Paramagentic and exhibits ionisation isomerism
Q	$[Ti(H_2O)_5Cl](NO_3)_2$	**2**	Diamagentic and exhibits *cis-trans* isomerism
R	$[Pt(en)(NH_3)Cl]NO_3$	**3**	Paramagentic and exhibits *cis-trans* isomerism
S	$[Co(NH_3)_4(NO_3)_2]NO_3$	**4**	Diamagentic and exhibits ionisation isomerism

Code :

	P	Q	R	S
(1)	4	2	3	1
(2)	3	1	4	2
(3)	2	1	3	4
(4)	1	3	4	2

89. Consider the following statements

S_1 : $[Cr(NH_3)_6]^{3+}$ is a inner orbital complex with crystal field stabilization energy equal to $-1.2\ \Delta_o$

S_2 : The complex formed by joining the CN^- ligands to Fe^{3+} ion has theoretical value of 'spin only' magnetic moment equal to 1.73 B.M.

S_3 : $Na_2S + Na_2[Fe(CN)_5NO] \longrightarrow Na_4[Fe(CN)_5NOS]$, In reactant and product the oxidation states of iron are same and arrange in the order of true/false.

(1) F T F (2) T T F

(3) T T T (4) F F F

90. In the isoelectronic series of metal carbonyl, the CO bond strength is expected to increase in the order:

(1) $[Mn(CO)_6]^+ < [Cr(CO)_6] < [V(CO)_6]^-$

(2) $[V(CO)_6]^- < [Cr(CO)_6] < [Mn(CO)_6]^+$

(3) $[V(CO)_6]^- < [Mn(CO)_6]^+ < [Cr(CO)_6]$

(4) $[Vr(CO)_6] < [Mn(CO)_6]^+ < [V(CO)_6]^-$

BIOLOGY

91. Which of the following is a pair of bacterial diseases?
(1) Typhoid and Pneumonia
(2) Malaria and AIDS
(3) Ringworm and AIDS
(4) Common cold and Malaria

92. Which is the most feared property of malignant tumors?
(1) Neoplasty
(2) Metastasis
(3) Rapid invasive growth
(4) Loss of contact inhibition

93. Given below are two statements
Statement I: Cancer detection is based on biopsy and histopathological studies of the tissue and blood and bone marrow tests for increased cell counts in the case of leukemias.
Statement II : The drugs, which are commonly abused are opioids, cannabinoids and coca alkaloids. Majority of these are obtained from flowering plants some are obtained from fungi.
Choose the correct answer from the option given below:
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

94. Which of the following cells are related with inflammatory reactions and allergic reactions
(1) Mast cells (2) Plasma cells
(3) Both (1) and (2) (4) T-lymphocytes

95. Given below are two statements
Statement I: Smoking is associated with increased incidence of cancers of lung, urinary bladder and throat, bronchitis, emphysema, coronary heart disease, gastric ulcer etc.
Statement II : The most common warming signs of drug and alcohol abuse among youth include drop in academic performance unexplained absence from school/collage, lack of interest in personal hygiene.
Choose the correct answer from the option given below:
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

96. A student when hear that he is failed in exam. after hearing this news he is mentally ill. His parents are goes with student to a doctor for this mental illness then doctor suggested which drug:
(1) Amphetamines
(2) Benzodiazepines
(3) Cocaine
(4) Both 1 and 2

97. Given below are two statements
Statement I: The human immune system consists of lymphoid organ, tissue, cells and soluble molecules like Antibodies.
Statement II : The immune system also plays an important role in allergic reaction, auto-immune disease and organ transplantations.
Choose the correct answer from the option given below:
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

98. Which of the following belongs to free living nitrogen fixing bacteria?
I. *Rhizobium* II. *Azospirillum*
III. *Azotobacter*
Choose the correct option
(1) I and II (2) I and III
(3) II and III (4) I, II and III

99. Which one of the following equipments is essentially required for growing microbes on a large scale, for industrial production of enzymes?
(1) Sludge digester (2) Industrial oven
(3) Bioreactor (4) BOD incubator

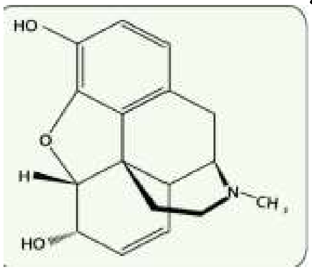
100. Which one of the following pair is not correctly matched?
(1) Filariasis – *Wuchereria* is the pathogen.
(2) Syphilis – Example of STD
(3) Common cold – Bacterial disease
(4) Dengue fever – Female *Aedes* is the vector

101. During an allergic reaction, the binding of antigens to IgE antibodies initiates a response, in which chemicals cause the dilation of blood vessels. Such chemicals are
(1) Interferons (2) Hormones
(3) Histamines (4) Acetyl choline

102. Select the wrong pair

(1)	Secondary treatment of sewage	Flocs
(2)	Primary treatment of sewage	Filteration and sedimentation
(3)	Anaerobic sludge diegesters	Biogas
(4)	Ganga action plan and Yamuna action plan	The ministry of energy and health

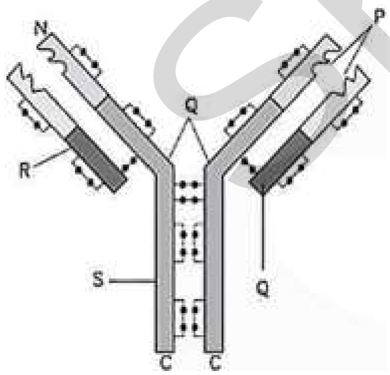
103. Select the correct statements for the given diagram



a. It is chemical structure of cocaine
b. It is a very effective sedative and painkiller
c. It is very useful in patients who have undergone surgery
d. It has a potent stimulating action on central nervous system
e. It is commonly called coke or crack is usually snorted

(1) b, c, e (2) Only b
(3) a, b, c, e (4) b, c

104. Refer to the given figure showing structure of an antibody. In the figure some parts are labelled as P, Q, R and S. Identify the part which binds with antigen.



(1) Q (2) P (3) R (4) S

105. Which of the following is correctly matched for the product produced by them?
(1) *Acetobacter aceti*–Antibiotics
(2) *Methanobacterium* – Lactic acid
(3) *Penicillum notatum* – Acetic acid
(4) *Saccharomyces cerevisiae* – Ethanol

106. Choose the incorrect pair.
(1) Lipases–Used in detergents for removing oil stains
(2) Pectinases and proteases–Used in clarifying bottled juices
(3) Statins–Competitively inhibit the enzyme responsible for cholesterol synthesis
(4) None of the above

107. Mark the option with correct statements about the lymph nodes
I. Are small solid structures along the lymphatic system.
II. Acts as filter for lymph or tissue fluid by trapping bacteria, viruses and other antigens, which are then destroyed by lymphocytes.

(1) Only I (2) Only II
(3) I and II (4) None of these

108. Select the incorrect statement for ringworm
(1) Many bacteria belonging to the genera *Microsporum, Trichophyton* and *Epidermophyton* are responsible for ringworms
(2) Ringworm are generally acquired from the soil or by using towels clothes or even the comb of infected individuals
(3) Appearance of dry, scaly lesions on various parts of the body such as skin, nails and scalp are the main symptoms of the disease
(4) It is one of the most common infectious disease in man

109. Which of the following is the correct increasing order of immunoglobulin percentage in human blood :
(1) IgD < IgE < IgM < IgA < IgG
(2) IgE < IgD < IgM < IgA < IgG
(3) IgG < IgA < IgM < IgD < IgE
(4) IgE < IgD < IgA < IgM < IgG

110. Treatment of snake bite by anti-venin is an example of :
(1) Artificially acquired active immunity
(2) Artificially acquired passive immunity
(3) Naturally acquired active immunity
(4) Naturally acquired passive immunity

111. During life cycle of *Plasmodium*, fertilization and development takes place in the mosquito___________.
(1) Blood (2) RBC's
(3) Gut (4) Liver

112. Given below are two statements :
Statement I: Health is Affected by genetic disorder, infection and life style.
Statement II: Health does not simply mean absence of disease or physical fitness.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct.

113. Given below are two statements :
Statement I: *Entamoeba histolytica* is a bacterial parasite in the large intestine of human.
Statement II: *Entamoeba histolytica* cause amoebic dysentery.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct.

114. Fill up the blanks.
I. Heat and moisture help ……fungi to grow, which makes them thrive in skin folds.
II. Maintenance of ……and…..hygiene is important for the prevention of many infectious diseases.
III. ……Gives the diseases its name. elephantiasis.
(1) I-*Microsporum*, II-public; personal, III- Swelling of hand
(2) I-*Trichophyton,* II-personal; public, III-Enlargement of eye
(3) I-*Epidermophyton*, II-personal; public, III-Swelling of legs
(4) I-*Wuchereria*, II-personal; public, III-Enlargement of tongue

115. Givenbelow are two statements
Statement I: Heroin is a depressant and slows down body functions.
Statement II: Cannabinoids are a group of chemicals which interact with cannabinoid receptors present principally in the brain.
Choose the correct answer from the option given below:
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

116. Select the correct match :

A. Tonsils	I. Primary lymphoid organs
B. HIV	II. Large reservoir of RBCs
C. Spleen	III. Retrovirus
D. Thymus	IV. Secondary lymphoid organ

(1) A –I, B – II, C – III, D – IV
(2) A –II, B – I, C – IV, D – III
(3) A –I, B – II, C – IV, D – III
(4) A –IV, B – III, C – II, D – I

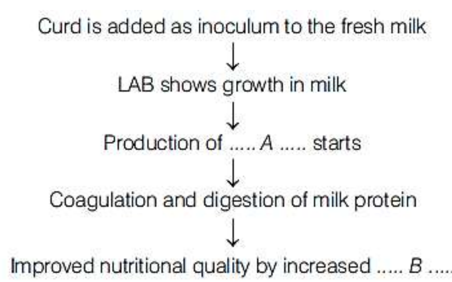
117. **Assertion (A):-** The chief component of biogas is CH_4.
Reason (R): Biogas plants are prepared on the foreign technology
(1) If both A and R are true and R is the correct explanation of A
(2) If both A and R are true, but R is not the correct explanation of A
(3) If A is true, but R is false
(4) If A is false, but R is true

118. Select the correct match :

A. Microsporum	I. Ring worm
B. Wuchereria	II. Filariasis
C. Virus	III. Common cold
D. Bacteria	IV. Plague

(1) A –I, B – II, C – III, D – IV
(2) A –IV, B – I, C – III, D – II
(3) A –III, B – I, C – IV, D – II
(4) A –IV, B – I, C – II, D – III

119. Study the following flowchart that shows curd formation from milk and select the correct option for *A* and *B*.



(1) A–citric acid, B–vitamin-B12
(2) A–lactic acid, B–vitamin-B12
(3) A–lactic acid, B–vitamin-C
(4) A–citric acid, B–vitamin-B2

120. Match the causative organisms with their diseases.

	Column-I		Column-II
(A)	*Haemophilus influenzae*	(i)	Malignant malaria
(B)	*Entamoeba histolytica*	(ii)	Elephantiasis
(C)	*Plasmodium falciparum*	(iii)	Pneumonia
(D)	*Wuchereria bancrofti*	(iv)	Typhoid
(E)	*Salmonella typhi*	(v)	Amoebiasis

(1) A-i, B-v, C-iii, D-ii, E-iv
(2) A-iii, B-v, C-i, D-ii, E-iv
(3) A-v, B-i, C-iii, D-iv, E-ii
(4) A-i, B-iii, C-ii, D-v, E-iv

121. Select the correct statements for AIDS
a. It is a type of viral disease
b. AIDS caused by HIV
c. AIDS was reported first time by 1981
d. It is always a type of congenital disease
(1) Only a, b (2) Only a, b, c
(3) Only b, c, d (4) All

122. Select the correct statement for Pneumonia
(1) *Haemophilus influenzae* are responsible for the disease pneumonia
(2) A healthy person acquires the infection by inhailing the droplets/aerosols released by an infected person or even by sharing glasses and utensils with an infected person
(3) In severe cases the lips and finger nails may turn gray in bluish in colour
(4) All of the above are true

123. Cancer cells sloughed from ______ tumors reach distant sites through blood and wherever they get lodged in the body, they start a new tumor there. This property called ____________ is the most feared property of malignant tumors.
(1) Malignant, contact inhibition
(2) Malignant, metastasis
(3) Benign, metastasis
(4) Benign, contact inhibition

124. With regard to the transmission of the HIV virus. Which one of the following statements is not correct?
(1) The chances of transmission from female to male are twice as likely as from male to female
(2) The chances of transmission are more through sexual route.
(3) An infected mother can transmit the infection to her baby during pregnancy.
(4) The risk of contracting infection from transfusion of infected blood is much higher
Than an exposure to contaminated needle

125 **Assertion (A):** Metastasis is the phenomenon in which tumour cells detach and migrate to other parts of the body where they give rise to secondary tumours.
Reason (R): All tumors show metastasis which is the most feared property.
(1) If both A and R are true and R is the correct explanation of A
(2) If both A and R are true, but R is not the correct explanation of A
(3) If A is true, but R is false
(4) If A is false, but R is true

126. Which of the following statements is/are correct?
(1) Biochemical Oxygen Demand (BOD) represents the amount of dissolved oxygen that would be consumed if all the organic matter in 1 L of water oxidised by microorganisms
(2) Sewage water is treated to reduce its BOD
(3) High value of BOD means the water is less polluted by organic matter
(4) Both (1) and (2)

127. Which of the following in which microbes are used
a. Production of biogas
b. Sewage treatment
c. Fermented beverages
d. Household products
e. Production of antibiotics
(1) Only a, b, c (2) Only b, c, e
(3) Only a, b, d (4) All

128. Select the Biofertilisers
Anabaena, Nostoc, Rhizobium,Azotobacter, TMV, Prions, Azospirillum.
(1) Five (2) Six (3) Four (4) Three

129. Given below are two statements
Statement I: Biofertilisers are organisms that enrich the nutrient quality of the soil. The main sources of biofertilisers are bacteria, fungi and cyanobacteria.
Statement II : The organic farmer holds the view that the eradication of the creatures that are often described as pests is not only possible, but also undesirable, for without them the beneficial predatory and parasitic insects which depend upon them as food or hosts would not be able to survive.
Choose the correct answer from the option given below:
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

130. Methanogens are found in
I. organic acid II. rumen of cattle
III. butanal IV. anaerobic sludge
Choose the correct option
(1) I and II (2) II and III
(3) II and IV (4) III and IV

131. Which of the following is wrongly matched in the given table ?

	Microbe	Product	Application
(1)	Streptococcus	Streptokinase	Removal of clotfrom blood vessel
(2)	Clostridium butylicum	Lipase	Removal of oil stains
(3)	Trichoderma polysporum	Cyclosporin A	Immuno suppressive drug
(4)	Monascus purpureus	Statins	Lowering of blood cholesterol

132. **Assertion(A):** Chemical pesticides are preferred over biopesticides.
Reason(R): Chemical pesticides are mostly expensive, hazardous and pollute the atmosphere.
(1) If both A and R are true and R is the correct explanation of A
(2) If both A and R are true, but R is not the correct explanation of A
(3) If A is true, but R is false
(4) If A is false, but R is true

133. Regarding common cold consider the following statements
I. Rhinovirus infects the nasal epithelium and respiratory passage but not the lungs
II. The symptoms of common cold included nasal congestion and discharge, sore throat, hoarseness, cough, headache and tiredness
Which of the statement given above is/are correct?
(1) Only I (2) Only II
(3) I and II (4) None of these

134. Match the disease in column I with the appropriate items in column II.

Column-I		Column-II	
(A)	Amoebiasis	(i)	STD
(B)	Diphtheria	(ii)	Mosquito bite
(C)	Dengue	(iii)	Bacterial disease
(D)	Hepatitis-B	(iv)	Protozoan Parasite

(1) A-(i), B-(ii), C-(iii), D-(iv)
(2) A-(ii), B-(iv), C-(i), D- (iii)
(3) A-(ii), B-(i), C-(iii), D- (iv)
(4) A-(iv), B-(iii), C-(ii), D-(i)

135. Which of the following forms the barriers of innate immunity?
I. Skin II. Macrophages
III.B-cells IV. Interferons
V. Antibodies VI. T-lymphocytes
VII. Fever
(1) I, II, IV and VII (2) II, III, V and VI
(3) III, V and VI (4) I, II, III, V and VII

136. Which of the following cells don't produce antibodies but stimates production of antibodies.
(1) T-Killer cells (2) T-Helper cells
(3) Plasma cells (4) Macrophages

137. Which of the following virus is not transferred through semen of an infected male
(1) Hepatitis-B virus
(2) Human immunodeficiency virus
(3) Chikungunya virus
(4) Ebola Virus

138. Roquefort cheese is formed by ripening with the fungi for a particular
(1) Colour (2) Flavor
(3) Shape (4) Texture

139. Given below are two statements , one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): All microbes are not harmful.
Reason (R): Microbes causes diseases in animals, plants and human beings but several microbes are useful to man in diverse ways.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(2) (A) is correct but (R) is not correct
(3) (A) is not correct but (R) is correct
(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

140. Given below are two statements
Statement I: Antibiotics produced by microbes are regarded as one of the most significant discoveries of the twentieth century and have greatly contributed towards the welfare of the human society.
Statement II : Statins acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.
Choose the correct answer from the option given below:
(1) Both Statement I and Statement II are incorrect
(2) Statement I is correct but Statement II is incorrect
(3) Statement I is incorrect but Statement II is correct
(4) Both Statement I and Statement II are correct

141. Identify the blank spaces *A, B, C* and *D* given in the following table and select the correct answer.

Types of microbes	Scientific names	Commercial products
Bacterium	*A*	Lactic acid
Fungus	B	Citric acid
C	*Acetobacter aceti*	Acetic acid
Fungus	*Penicillium notatum*	*D*

(1) A–*Lactobacillus*, B–*Aspergillus niger*, C–Bacterium, D–Penicillin
(2) A–*Staphylococcus*, B–*Clostridium, C*–Yeast, *D*–Penicillin
(3) A–*Lactobacillus*, B–*Microsporum*, C–Yeast, D–Penicillin
(4) A–*Staphylococcus*, B–*Microsporum*, C–*Agaricus*, D–Penicillin

142. Cancer detection is based on ______.
(a) Biopsy and histopathological studies of the tissues
(b) Blood and bone marrow tests for increased cell counts in the case of leukemias
(c) Radiography, computed tomography and magnetic resonance imaging
(1) a and b (2) b and c
(3) a and c (4) a, b and c

143. Cyclosporin is the drugs that is used as ………. after organ transplantation.
(1) Anti retroviral drugs
(2) Immuno-suppressants
(3) Immuno-modulators
(4) Immuno-vaccines

144. Sewage contains large amounts of …*A*… and …*B*…. Here *A* and *B* refer to
(1) A–inorganic matter, B–bacteria
(2) A–organic matter, B–pathogenic microbes
(3) A–organic matter, B–virus
(4) A–inorganic matter, B–pathogenic microbes

145. Match the following columns.

Column-I (Organisms)		Column-II (Uses)	
(A)	*Lactobacillus*	(i)	Penicillin
(B)	*Saccharomyces cerevisiae*	(ii)	Swiss cheese
(C)	*Propionibacterium shermanii*	(iii)	Bread
(D)	*Penicillium*	(iv)	Milk into curd

(1) A-iv, B-iii, C-ii, D-i
(2) A-iii, B-ii, C-i, D-iv
(3) A-iv, B-i, C-ii, D-iii
(4) A-i, B-iv, C-iii, D-ii

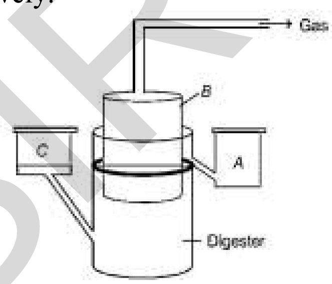
146. Match the column-I and column-II.

Column-I		Column-II	
(a)	Dragon flies	(i)	Rid of aphids
(b)	Lady bird	(ii)	Attack on insects and other Arthropods
(c)	*Glomus*	(iii)	Mycorrhiza
(d)	Baculoviruses	(iv)	Rid of mosquitoes

(1) a-iv, b-i, c-iii, d-ii (2) a-i, b-iv, c-ii, d-iii
(3) a-ii, b-i, c-iv, d-iii (4) a-iv, b-i, c-ii, d-iii

147. Which one of the following in sewage treatment removes suspended solids?
(1) Tertiary treatment (2) Secondary treatment
(3) Primary treatment (4) Sludge treatment

148. The diagram given below represents a typical biogas plant. Select the correct option for A, B and C, respectively.



(1) A-Sludge, B− Dung + Water, C− Gas holder
(2) A-Dung + Water, B-Sludge, C−CH_4 + CO_2
(3) A-Sludge, B - Gas holder, C - Dung + Water
(4) A-CH_4 + CO_2, B - Dung + Water, C− Sludge

149. Choose the correct option to fill in the blanks.
I. Primary treatment of sewage involves physical removal of small and large particles through ...*A*... and ...*B*... .
II. Microbes consume the major part of organic matter in the effluent and reduce ...*C*... of sewage.
III. ...*D*... particularly ...*E*... anaerobically breakdown cellulosic material and produce CO_2 and H_2 from anaerobic sludge during ...*F*... . treatment.
IV. When BOD of sewage has reduced, the effluent is passed into ...*G*... . Here, *A*to*G*can be
(1) A–sedimentation, B–centrifugation, C–BOD, D–Methanogen, E–*Methanobacterium*, F–water gas plant, G–settling tank
(2) A–centrifugation, B–sedimentation, C–BOD, D–Methanogen, E–*Methanococcus*, F–biogas plant, G–sludge tank
(3) A–filtration, B–centrifugation, C–BOD, D–Methanogen, E–*Methanobacillus*, F–gobar gas plant, G–filtre tank
(4) A–filtration, B–sedimentation, C–BOD, D–Methanogen, E–*Methanobacterium*, F–sewage treatment plant, G–settling tank

150. Consider the following statements
Statement I- Neural system and endocrine system influences our immune system
Statement II- Immune system maintains our health
(1) Statement I is correct and statement II is incorrect
(2) Statement I is incorrect and statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

151. *Salmonella typhi* are bacteria, causes typhoid, pathogen cause lesions and ulceration in the ______. There is a pea-soup diarrhoea which may become haemorrhagic.
(1) Stomach wall (2) Intestinal wall
(3) Gall bladder (4) Blood vessels

152. Regarding pathogens consider the following statements
Statement-I: A pathogen or an infectious agent is a microorganism, such as a virus, bacterium, fungus that causes disease in its host
Statement-II: These pathogens multiply in our body and interfere with the normal vital activities, resulting in morphological and functional damage
(1) Statement I is correct and statement II is incorrect
(2) Statement I is incorrect and statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

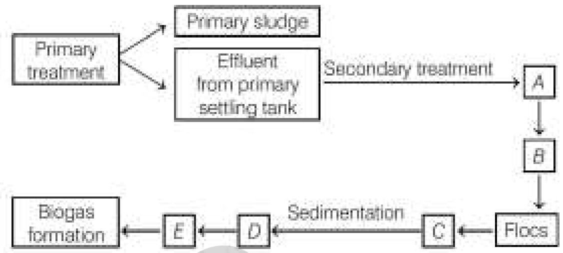
153. Mycorrhiza is a symbiotic fungal association with roots of higher plants. How is it beneficial to the plant:
(1) Mycorrhiza provides phosphorus to the plants from the soil.
(2) Make the plant resistant to root-borne pathogen.
(3) Increase tolerance to salinity and drought.
(4) All of the above

154. Side effects of anabolic steroids in females include
I. Masculinisation.
II. Aggressiveness.
III. Mood swings, depression.
IV. Abnormal menstrual cycle.
V. Excessive facial and body hair.
(1) I, II and III (2) I, II, III and IV
(3) II, III, IV and V (4) I, II, III, IV and V

155. Pollution from animal excreta and organic waste from kitchen can be most profitably minimised by
(1) Storing them in underground storage tanks
(2) Using them for producing biogas
(3) Vermiculture
(4) Using them directly as biofertilizers

156. Inflammatory response in allergy is due to the release of …… from the mast cells
(1) Antigen (2) Antibody
(3) Histamine (4) None of these

157. Given below is the flowchart of sewage treatment. Identify *A, B, C, D* and *E* and select the correct option.



(1) A–Small aeration tank, B–Microbial digestion, C–High BOD, D–Activated sludge, E–Aerobic sludge digesters
(2) A–Large aeration tank, B–Mechanical agitation, C–Increased BOD, D–Activated sludge, E–Aerobic sludge digesters
(3) A–Small aeration tank, B–Microbial digestion, C–Low BOD, D–Activated sludge, E–Anaerobic sludge digesters
(4) A–Large aeration tank, B–Mechanical agitation, C–Reduced BOD, D–Activated sludge, E–Anaerobic sludge digesters

158. **Assertion:** Besides curdling of milk, LAB also improve its nutritional quality by increasing vitamin-B_{12}
Reason : LAB, when present in human stomach, check disease causing microbes
(1) If both assertion and reason is true and reason is the correct explanation of assertion.
(2) If both assertion and reason is true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If assertion is false but reason is true.

159. Match the column-I and column-II.

	Column-I		Column-II
(a)	Whisky, brandy and rum	(i)	Without distillation
(b)	Ernest Chain and Howard Florey	(ii)	Brewer's yeast
(c)	*Saccharomyces cerevisiae*	(iii)	Full potential as an effective antibiotic (penicillin) was established
(d)	Wine and beer	(iv)	distillation

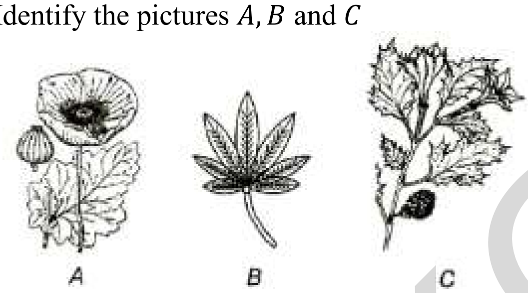
(1) a-iv, b-iii, c-ii, d-i
(2) a-i, b-iii, c-ii, d-iv
(3) a-ii, b-i, c-iv, d-iii
(4) a-iv, b-iii, c-i, d-ii

160. Which of these is incorrectly matched?

	Disease	Pathogen	Causative agent
(1)	Typhoid	*Salmonella*	Bacteria
(2)	Filariasis	*Microsporum*	Virus
(3)	Ascariasis	*Ascaris*	Round worm
(4)	Common cold	Rhinovirus	Virus

161. Biopesticides are
(1) The chemicals which are used to destroy the pests
(2) The living organism or their products which are used for pest control
(3) The organisms which destroy the crops
(4) None of these

162. Identify the pictures A, B and C



(1) A-Opium poppy, B-*Cannabis sativa*, C-*Datura*
(2) A-*Cannabis sativa*, B-Opium poppy, C-*Datura*
(3) A-*Datura*, B-Opium poppy, C-*Cannabis sativa*
(4) A-Opium poppy, B-*Datura*, C-*Cannabis sativa*

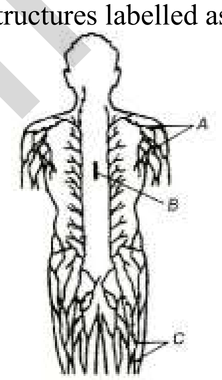
163. Penicillin was discovered by chance from *Penicillium notatum*, which was seen to grow in an unwashed culture plate of *Staphylococci*. Staphylococci is a :
(1) Bacteria (2) Algae
(3) Fungus (4) Virus

164. Microbes are used in
I. primary treatment of sewage.
II. secondary treatment of sewage.
III. anaerobic sludge digesters.
IV. production of biogas.
Choose the correct option.
(1) I, II and III (2) I, III and IV
(3) II, III and IV (4) All of the above

165. Smack is chemically …A… which is white and odourless and crystalline in nature. This is obtained by …B…. Here A and B refers to
(1) A-diacetyl morphine; B-acetylation of morphine
(2) A-morphine; B-acetylation of hashish
(3) A-stimulant; B-acetylation of morphine
(4) A-hallucinogen; B-acetylation of hashish

166. Which one of the following statements is correct with respect to immunity?
(1) Preformed antibodies need to be injected to treat the bite by a viper snake
(2) The antibodies against smallpox pathogen are produced by T-lymphocytes
(3) Antibodies are protein molecules, each of which has four light chains
(4) Rejection of a kidney graft is the function of B-lymphocytes

167. Identify the structures labelled as A, B and C



(1) A-Lymph nodes, B-Thymus, C-Lymphatic vessels
(2) A-Lymphatic vessels, B-Thyroid, C-Lymph nodes
(3) A-Tonsils, B-Peyer's patchs, C-Lymphatic vessels
(4) A-Tonsils, B-Thymus, C-Peyer's patches

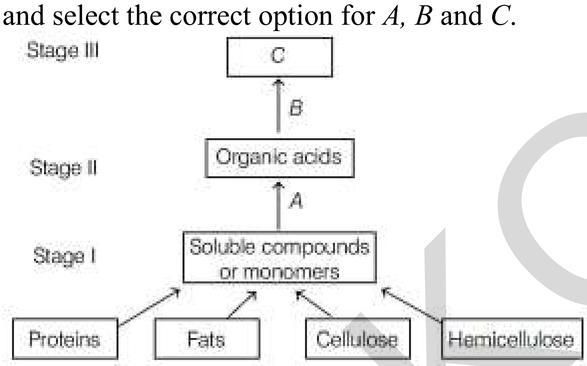
168. The BOD test measures the rate of uptake of oxygen by microbes in water bodies. The greater BOD of sample water, indicates that
(1) it is highly polluted
(2) it is not polluted
(3) it is moderately polluted
(4) pollution level cannot be determined

169. Which group of bacteria is used for the treatment of waste water in STPs :
(1) Autotrophs (2) Heterotrophs
(3) Cyanobacteria (4) Both 1 and 2

170. Select the correct statement from the set given below:
Statement I- Drugs like barbiturates and amphetamines are being used for treatment of mental illness.
Statement II- Several plants, fruits and seed having hallucinogenic properties have been used in folk-medicine
(1) Statement I is correct and statement II is incorrect
(2) Statement I is incorrect and statement II is correct
(3) Both Statement I and Statement II are correct
(4) Both Statement I and Statement II are incorrect

171. **Statement I:** Histamine is released during allergic reactions.
Statement II: Histamine dilates blood vessels and increases vascular permeability.
(1) Both I and II are correct, and II is the correct explanation of I
(2) Both I and II are correct, but II is not the correct explanation of I
(3) I is correct, II is incorrect
(4) I is incorrect, II is correct

172. Study the following flowchart of biogas production and select the correct option for *A, B* and *C*.



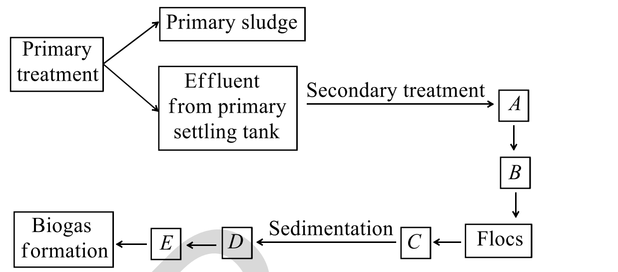
(1) A–Methanogenic bacteria, B–Fermentative microbes, C–CO_2 and hydrogen (biogas)
(2) A–Anaerobic microorganisms, B–*Methanococcus*, C–CO_2 and nitrogen (biogas)
(3) A–Anaerobic microbes, B–Methanogenic bacteria, C–CO_2 and methane (biogas)
(4) A–Aerobic microorganism, B–*Methanobacter*, C–CO_2 and methane (biogas)

173. Mark the correct option regarding A, B, C and D is

Diseases	Causative Organisms	Symptoms
I. Filariasis	*A*	Chronic inflammation
II. Typhoid	B	Sustained high fever, stomach pain
III. *C*	Rhinoviruses	Nasal congestion and discharge
IV. Ascariasis	*Ascaris*	D

(1) A-*Wuchereria*, D-Internal bleeding, fever, anaemia
(2) B-*Ascaris*, C-Typhoid,
(3) B-*Entamoeba histolytica*, D-Constipation, fever
(4) A-*Entamoeba histolytica*, B- *Salmonella typhi*,

174. Given below is the flowchart of sewage treatment. Identify A, B, C, D and E and select the correct option



(1) A-small aeration tank, B-Microbial digestion, C-High BOD, D-Activated sludge, E-Aerobicsludge digesters
(2) A-Large aeration tank, B-Mechanical agitation, C-Increased BOD, D-Activated sludge, E-Aerobic sludge digesters
(3) A-small aeration tank, B-Microbial digestion, C-Low BOD, D-Activated sludge, E-Anaerobic sludge digesters
(4) A-Large aeration tank, B-Mechanical agitation, C-Reduced BOD, D-Activated sludge, E-Anaerobic sludge digesters

175. Consider the following statements. Which of the statements given below is incorrect?
(1) Microbes are also used to ferment fish, soyabean and bamboo shoots to make food
(2) Different varieties of cheese are known by their characteristic texture, flavour and taste, the specificity coming from the microbe used
(3) *Trichoderma* sp., free living fungi, are present in root ecosystem where there act against several plant pathogen
(4) *Rhizobium* is a symbiotic bacterium that lives in the stem of legumes

176. Consider the following statements
I. People should get vaccination to avoid infection
II. Vaccination is available against polio, cholera, typhoid, tuberculosis and many other disease
III. Eradication of vectors are necessary in diseases like malaria and filariasis
IV. Dengue and chikungunya, both are spread by *Culex* mosquitoes
Which of the statements given above are correct?
(1) I, II and III
(2) I, II and IV
(3) I, III and IV
(4) III and IV

177. **Assertion:** A person who has received a cut and is bleeding needs to given anti-tetanus treatment.
Reason: ATS provides active immunity by stimulating production of antibodies.
(1) If both assertion and reason is true and reason is the correct explanation of assertion.
(2) If both assertion and reason is true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If assertion is false but reason is true.

178. Match the following columns and choose the correct option.

Column-I		**Column-II**	
(A)	Protozoan disease	(i)	Ring worm
(B)	Helminthal disease	(ii)	Amoebiasis
(C)	Bacterial disease	(iii)	Plague
(D)	Fungal disease	(iv)	Filariasis

(1) A-ii, B-iv, C-iii, D-i
(2) A-i, B-ii, C-iii, D-i
(3) A-ii, B-i, C-iii, D-iv
(4) A-iii, B-i, C-iv, D-ii

179. Which of the following is/are the approache(s) for biological farming?
I. Familiarity with various life forms inhabiting the field.
II. Gain knowledge about the life cycles, patterns of feeding and habitat of predators and pests.
Choose the correct option.
(1) Only I
(2) Only II
(3) I and II
(4) None of these

180. Select the true statement
A. AIDS has no cure, prevention is the best option
B. During HIV infection, the person suffers from fever, weight loss and diarrhoea
C. HIV is a retrovirus, which means its genetic material is DNA which can produce RNA.
(1) A and C
(2) B and C
(3) A and B
(4) A, B and C

Answer Key
and
Solution

ANSWER KEY & SOLUTION

Que.	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Ans.	3	2	4	3	2	2	1	2	2	2	2	3	1	2	2
Que.	**16**	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
Ans.	3	1	2	2	1	2	3	1	3	1	3	2	4	3	2
Que.	**31**	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
Ans.	1	2	2	1	3	2	3	4	1	1	1	4	3	1	4
Que.	**46**	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
Ans.	3	4	3	4	3	3	2	4	1	2	3	1	2	4	1
Que.	**61**	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**	**71**	**72**	**73**	**74**	**75**
Ans.	2	1	2	3	1	2	2	4	2	4	2	4	3	2	2
Que.	**76**	**77**	**78**	**79**	**80**	**81**	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
Ans.	1	3	3	2	3	4	1	1	4	2	2	1	2	3	2
Que.	**91**	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**	**101**	**102**	**103**	**104**	**105**
Ans.	1	2	4	1	4	4	4	3	3	3	3	4	4	2	4
Que.	**106**	**107**	**108**	**109**	**110**	**111**	**112**	**113**	**114**	**115**	**116**	**117**	**118**	**119**	**120**
Ans.	4	3	1	2	2	3	4	3	3	4	4	3	1	2	2
Que.	**121**	**122**	**123**	**124**	**125**	**126**	**127**	**128**	**129**	**130**	**131**	**132**	**133**	**134**	**135**
Ans.	2	4	2	1	3	4	4	1	4	3	2	4	3	4	1
Que.	**136**	**137**	**138**	**139**	**140**	**141**	**142**	**143**	**144**	**145**	**146**	**147**	**148**	**149**	**150**
Ans.	2	3	2	4	4	1	4	2	2	1	1	3	3	4	3
Que.	**151**	**152**	**153**	**154**	**155**	**156**	**157**	**158**	**159**	**160**	**161**	**162**	**163**	**164**	**165**
Ans.	2	3	4	4	2	3	4	2	1	2	2	1	1	3	1
Que.	**166**	**167**	**168**	**169**	**170**	**171**	**172**	**173**	**174**	**175**	**176**	**177**	**178**	**179**	**180**
Ans.	1	1	1	2	3	1	3	1	4	4	1	3	1	3	3

PHYSICS

1. **(3)**

Sol. A transformer changes voltage and current but the power remains constant.

2. **(2)**

Sol. Current due to orbital motion of electron I = eυ.
Magnetic dipole moment,
$M = AI = (\pi r^2)(e\upsilon) = \pi r^2 \upsilon e$

3. **(4)**

Sol. By using $\tan\phi = \frac{B_V}{B_H} \Rightarrow \tan 60^o = \frac{B_V}{0.38\times10^{-4}}$

$\Rightarrow B_V = 0.38\times10^{-4}\times\sqrt{3} = 0.62\times10^{-4}$.

4. **(3)**

Sol. By using $\tan\phi' = \frac{\tan\phi}{\cos\beta}$; where $\phi = 40^o, \beta = 30^o$

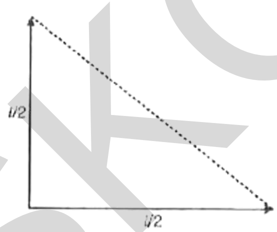
As $\cos 30^o < 1$ $\Rightarrow \frac{1}{\cos 30^o} > 1$

Hence $\frac{\tan\phi'}{\tan\phi} > 1 \Rightarrow \tan\phi' > \tan\phi = \phi' > \phi$

or $\phi' > 40^o$

5 **(2)**

Sol. Magnetic moment of each part is $= M/2$
So, the net magnetic moment is



$= \sqrt{\left(\frac{M}{2}\right)^2 + \left(\frac{M}{2}\right)^2} = \frac{M}{\sqrt{2}}$

6. **(2)**

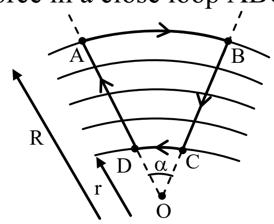
Sol. When a magnet is freely suspended in earth's magnetic field, it's north pole points north, so the magnetic field of the earth may be suppose to be due to a magnetic dipole with it's south pole towards north and as equatorial point is on the broad side on position of the dipole.

$B_e = \frac{\mu_0}{4\pi}.\frac{M}{r^3} \Rightarrow 0.3\times10^{-4} = 10^{-7}\times\frac{M}{(6.4\times\times10^6)^3}$

$\Rightarrow M = 7.8\times10^{22} A\text{-}m^2$.

7. **(1)**

Sol. To prove this we will calculate work done by electric force in a close loop ABCD.



during displacement AD and BC work done by electric force = 0 as electrostatic is perpendicular to displacement.
Work done during AB = R α E_1
Work done during CD = – r α E_2
Total work done = R α E_1 – r α E_2
Close loop work done should be zero

$\therefore \frac{E_1}{E_2} = \frac{r}{R}$

8. **(2)**

Sol. Paramagnetic substance moves with weak force from weak magnetic field to strong magnetic field.

9. **(2)**

Sol. Here effective length is 2R

$\varepsilon = \frac{1}{2}B\omega(2R)^2 = 2B\omega R^2$

10. **(2)**

Sol. For paramagnetic material,

$\chi \propto \frac{1}{T} \Rightarrow \chi_1 T_1 = \chi_2 T_2$

$\frac{6}{0.4}\times 4 = \frac{M}{0.3}\times 24$

$I = \frac{0.3}{0.4} = 0.75\ A/m$

11. **(2)**

Sol. An electric moving around the nucleus has a magnetic moment μ_r given by

$\mu_l = \frac{e}{2m}L$

Where L is the magnitude of the angular momentum of the circulating electron around the nucleus. The smallest value of μ_r is called the Bohr magneton μ_B and its value is $\mu_B = 9.27\times10^{-24}\ JT^{-1}$

12. **(3)**

Sol. Diamagnetic substances art those substances in which resultant magnetic moment in an atom is zero

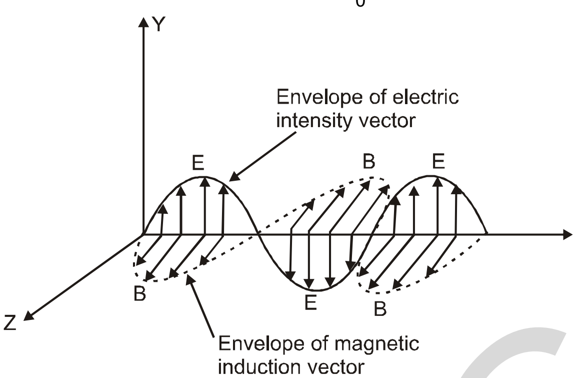
A paramagnetic material tends to move from a weak magnetic field to strong magnetic field.

A magnetic material is in the paramagnetic phase above its Curie temperature.

Typical domain size of a ferromagnetic material is 1mm

13. **(1)**

Sol. The amplitudes of the electric and magnetic fields in free space are related by $\frac{E_0}{B_0} = c$



In figure, electric field vector $(\vec{E})$ and magnetic field vector $(\vec{B})$ are vibrating along Y and Z directions and propagation of electromagnetic wave is shown in X-direction.

Hence, electric and magnetic fields are in phase and perpendicular to each other.

14. **(2)**

Sol. $\omega = 2\pi v = \frac{2\pi c}{\lambda} = \frac{2\pi \times 3 \times 10^8}{6 \times 10^{-3}}$

$= \pi \times 10^{11} \text{rad/sec}$

$E_y = E_0 \sin \omega t(t - x/c)$

$= 33 \sin \pi \times 10^{11}(t - x/c)$

15. **(2)**

Sol. The correct relation is $\mu_r = 1 + \chi$

16. **(3)**

Sol. Self inductance $= \frac{\mu_0 N^2 A}{l}$

17. **(1)**

Sol. Electric energy stored in capacitor

$U = \frac{1}{2}\frac{q}{C} = \frac{1}{2}\frac{q^2}{\varepsilon_0 A/d}$

Multiply and divide by A

$= \frac{1}{2}\frac{q^2}{\varepsilon_0 A^2} \times A \times d = \frac{q^2 d}{2\varepsilon_0 A} \text{ or } \frac{U}{Ad} = \frac{1}{2}\frac{q^2}{\varepsilon_0 A^2}$

18. **(2)**

Sol. $L_1 \frac{di_1}{dt} = L_2 \frac{di_2}{dt}$

or $L_1 di_1 = L_2 di_2$ or $L_1 i_1 = L_2 i_2$

$\therefore \frac{i_1}{i_2} = \frac{L_2}{L_1}$

19. **(2)**

Sol. Hysteresis is shown by the ferromagnetic material

20. **(1)**

Sol. $\chi_d < \chi_p < \chi_f$

21. **(2)**

Sol. The liquid rises up in the part of the tube which is between the poles.

22. **(3)**

Sol. $n = \frac{x}{\lambda} = \frac{10^{-3}}{4 \times 10^{-7}} = 0.25 \times 10^4$

23. **(1)**

Sol. The net magnetic flux through a closed surface. will be zero. $\oint \vec{B}.\vec{dS}$ because there are no magnetic monopoles.

24. **(3)**

Sol. $|\varepsilon| = L\frac{dI}{dt} = \frac{0.5(10-0)}{2} = 2.5 \text{volt}$

25. **(1)**

Sol. Electromagnetic waves transfer energy & momentum on hitting the object or material.

26. **(3)**

Sol. $\tan\phi = \frac{X}{R} = 1 \Rightarrow \phi = \tan^{-1}(1) = 45^0$

Power factor $= \cos\phi = \cos 45^0 = \frac{1}{\sqrt{2}}$

27. **(2)**

Sol. Here, V = 220 V

Resistance, R = 22 Ω

Impedance, Z = 44 Ω

Current in circuit, $I = \frac{V}{Z} = \frac{200V}{44\Omega} = 5A$

Power dissipated in the circuit,

$P = I^2R = (5)^2 \times 22 = 550$ w

28. **(4)**

Sol. Impedance of the LCR circuit,

$Z = \sqrt{R^2 + (X_L - X_C)^2}$

At resonance, $X_L = X_C$

$\therefore Z = R$

Power factor, $\cos\phi = \frac{R}{Z} = 1$

29. **(3)**

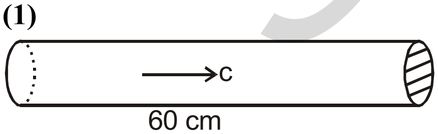
Sol.

30. **(2)**

Sol.

31. **(1)**

Sol.



The time taken by the electaromagnetic wave to move through a distnce of 60 cm is t = $\frac{60 \text{ cm}}{c}$ = 2×10^{-9} s.The energy contained in the 60 cm length passes through a cross-section of the beam in 2×10^{-9} s. But the energy passing through any cross-section in 2×10^{-9} s is

$U = (4 \text{ mW}) \times (2 \times 10^{-9} \text{ s})$

$= (4 \times 10^{-3} \text{ J/s}) \times (2 \times 10^{-9} \text{ s}) = 8 \times 10^{-12}$ J.

This is the energy contained in 60 cm length.

32. **(2)**

Sol. $\mu_0 H = \mu n i$

$H = ni = \left(\frac{100}{0.2}\right) \times 5.2 = 2600$ A/m

33. **(2)**

Sol. (p) → (i) , (q) → (ii), (r) → (iii), s → (iv)

34. **(1)**

Sol. $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$

35. **(3)**

Sol. $C = \frac{E_0}{B_0}$

$\therefore \quad B_0 = \frac{E_0}{C} = 1.6 \times 10^{-6}$ Wb/m²

36. **(2)**

Sol. $\mu = \frac{c}{v} = \frac{1}{\sqrt{\mu_0\varepsilon_0}} \times \frac{\sqrt{\mu\varepsilon}}{1} = \sqrt{\frac{\mu\varepsilon}{\mu_0\varepsilon_0}}$

37. **(3)**

Sol. Since the magnetic field is uniform, hence there is no change in the magnetic flux. Linked with cylindrical wire and hence no current will be induced on the surface of cylindrical wire.

38. **(4)**

Sol. $\xrightarrow[\lambda \text{ increasing}]{\text{Y X U V I M R}}$

Hence energy of radio wave will be minimum and maximum for X - ray

39. **(1)**

Sol. When dc is applied $i = \frac{V}{R} \Rightarrow 1 = \frac{100}{R} \Rightarrow$ R = 100Ω. When ac is applied $i = \frac{V}{Z} \Rightarrow 0.5 = \frac{100}{Z} \Rightarrow$ Z = 200Ω. Hence

$Z = \sqrt{R^2 + X_L^2} = \sqrt{R^2 + 4\pi^2\nu^2L^2} \Rightarrow$

$(200)^2 = (100)^2 + 4\pi^2(50)^2 L^2 \Rightarrow$ L = 0.55H.

40. **(1)**

Sol. Impedance of the circuit = $\sqrt{R^2 + \omega^2 L^2}$

Amplitude of voltage $= V_o$

$\therefore$ Amplitude of current $= \dfrac{V_o}{\sqrt{R^2 + \omega^2 L^2}}$,

Hence (1) is correct.

41. **(1)**

Sol. $\dfrac{1}{L_{eq}} = \dfrac{1}{L} + \dfrac{1}{L} + \dfrac{1}{L} = \dfrac{3}{L} \Rightarrow L_{eq} = \dfrac{L}{3}$

42. **(4)**

Sol. X – ray Crystallography is based on diffraction principle

43. **(3)**

Frequency remains unchanged in a transformer frequency at input = Frequency at output

44. **(1)**

Sol. Wavelength of microwave is more than that of infrared, it ranges from 10^{-3} to 0.3 m and that for infrared wavelength it ranges from 8×10^{-7}m – 10^{-3}m.

45. **(4)**

Sol. Radio waves

CHEMISTRY

46. **(3)**

Sol. Fe has +1 oxidation state in nitroprusside ion
$(x + 5(0) + 1 = +2 \Rightarrow x = +1)$

47. **(4)**

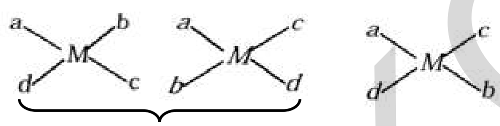
Sol. 25% $\left(\frac{1^{th}}{4}\right)$ mole of 4Cl atoms = 1
& in $ptCl_4.3NH_3$ with pt having o.5 = +4
(Coordination number = 6)
$\therefore$ using Werner theory.
The compound will be $[pt\,(NH_3)_3\,Cl_3]\,Cl$

48. **(3)**

Sol. Ni^{2+} ion is identified by Dimethyl glyoxime
(dmg^-) (rosy – red ppt.)
$Ni^{2+} + 2\,dmg \xrightarrow{NH_4OH} [Ni\,(dmg)_2]$
Rosy Red ppt

49. **(4)**

Sol. [Mabcd] gives three Geometrical isomerism
(2 cis & one trans)



cis(w.r.t a and b) Trans

50. **(3)**

Sol. $K_4\,[Fe(CN)_6]s$
$Fe^{2+} \Rightarrow [Ar]3d^6$
CN $\Rightarrow$ SFL $\Rightarrow$ pairing occurs

$\rightarrow$Zero unpaired
$\rightarrow$diamagnetic

Option (3) is wrong

51. **(3)**

Sol. $[Cr(en)_3]^{3+}$ shows no Geometrical isomerism
($[M(AA)_3]$ type)

52. **(2)**

Sol. $CoCl_3.3NH_3 \Rightarrow [Co(NH_3)_3\,Cl_3]$ gives only one ion
$\therefore$ it has least conductivity in aq. solution.

53. **(4)**

Sol. Zeise's salt $K\,[Pt\,(C_2H_4)\,Cl_3]$ has more than one type of ligand: It is heteroleptic.

54. **(1)**

Sol. $2MnO_4^- + H_2O + I^- \longrightarrow 2MnO_2 + 2OH^- + IO_3^-$

55. **(2)**

Sol. $[Rh(III)(en)_2(ONO)\,(SCN)]^+\,(NO_3^-)$

56. **(3)**

Sol. EAN of Ni - 28 - 2 + 8 = 34
EAN of Cu = 29 - 2 + 8 = 35
EAN of Pt = 78 - 4 + 12 = 86

57. **(1)**

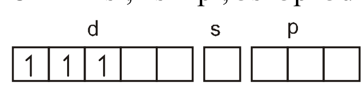
Sol. On the basis of number of electrons the correct order is P > Q > R > S.

58. **(2)**

Sol. (1) Exists as Al^{3+} and C^{4-}
(2) Tetraethyl lead is sigma bonded organometallic compound

59. **(4)**

Sol. Atoms, ions or molecules having unpaired electrons are paramagnetic. In $[Cr(NH_3)_6]^{3+}$, Cr is present as Cr (III).
$Cr^{3+} = 1s^2, 2s^2\,2p^6, 3s^2\,3p^6\,3d^3$



d^2sp^3

60. **(1)**

Sol. On charge balancing, $[Co(III)(NH_3)_5(CO_3)]^+ + Cl^-$.

61. **(2)**

Sol.

62. **(1)**

Sol. In $K_2Cr_2O_7$, the oxidation number of Cr = +6

The electronic configuration of Cr^{6+} is

$_{24}C^{6+} \rightarrow [Ar]3d^0 4s^0$

So it has no unpaired electron but is orange coloured.

63. **(2)**

Sol. +3 oxidation state is common for all lanthanoids

64. **(3)**

Sol. $Cr_2O_7^{2-} + 14H^+ + 3Sn^{2+} \rightarrow 2Cr^{3+} + 3Sn^{4+} + 7H_2O$

65. **(1)**

Sol. The transition metals and their compounds are known for their catalytic activity because of their ability to adopt multiple oxidation states and to form complexes.

66. **(2)**

Sol. $Fe^{2+} - 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$

$Fe^{3+} - 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$

In Fe^{3+}, orbital in half filled hence provides extra stability.

67. **(2)**

Sol. Large value of second ionization enthalpy of copper is compensated by much more negative hydration energy of $Cu^{2+}_{(aq)}$

68. **(4)**

Sol.

$$K_2Cr_2O_7 + 7H_2SO_4 + 6KI \rightarrow 4K_2SO_4 + Cr_2(SO_4)_3 + 3I_2 + 7H_2O$$

69. **(2)**

Sol. In case of transition metal oxides, the oxides with metals in lower oxidation states are basic in nature.

O.S. of Mn in

$Mn_2O_7 = +7; V \text{ in } V_2O_3 = +3; V$ in $V_2O_5 = +5$; cr in CrO

$= +2; Cr \text{ in } Cr_2O_3 = +3$

Thus in V_2O_3, CrO and $Cr_2O_3 = +3$ thus in V_2O_3 CrO and Cr_2O_3 is amphoteric in nature hence V_2O_3 and CrO are basic in nature

70. **(4)**

Sol. (A) → (iii), (B) → (i), (C) → (ii), (D) → (iv), (E) → (v)

71. **(2)**

Sol. (A) → (ii), (B) → (iii), (C) → (iv), (D) → (i)

72. **(4)**

Sol. Strong metallic bonds between the atoms of transition elements attribute to their high melting and boiling points. Zinc has all electrons paired ([Ar] $3d^{10}$ $4s^2$) and thus do not participate in metallic bonding. So accordingly its melting point is least.

73. **(3)**

Sol. $[Ni(CO)_4]$; Ni is in zero oxidation state. CO acts as σ doner as well as π acceptor.

74. **(2)**

Sol.

75. **(2)**

Sol.

76. **(1)**

Sol. No. of unpaired electrons are:'

$Ti^{3+} = 1, Cu^+ = 0, Zn^{2+} = 0, Sc^{3+} = 0$

77. **(3)**

Sol. $[Xe]4f^{14}5\,d^{10}6\,s^1$ among the given electronic configurations is not the configuration of lanthanoid.

78. **(3)**

Sol. Multidentate ligand can cause chelation.

79. **(2)**

Sol. EAN = 28 – 2 + 8 = 34

80. **(3)**

Sol. In $Ni(CO)_4$, all electrons are paired

81. **(4)**

Sol. Trans complex is symmetrical and is optically inactive.

82. **(1)**

Sol. $[Ni(NH_3)_6]^{2+}$ has sp^3d^2 hybridisation having octahedral geometry as with d^8 configuration no two empty d-orbitals are available for d^2sp^3 hybridisation. As sp^3d^2 hybridisation involves nd orbital (i.e. outer orbitals), so the complex is called as outer orbital complex.

83. (1)

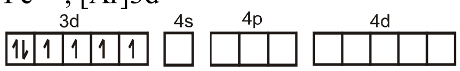
Sol. Let x be the oxidation state of cobalt in $[CoCl_2(en)_2]^+$. So, x + 2(–1) + 0 = +1 or x = + 3.

As there are six σ-bonds between cobalt and ligands, it coordination number is 6 (here 'en' is a bidentate ligand).

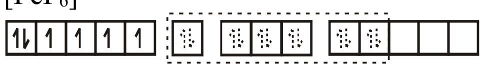
84. (4)

Sol. The complex $[fef_6]^{4-}$ is paramagnetic and uses outer orbital (4d) in hybridisation (sp^3d^2) ; it is thus called as outer orbital or high spin or spin free complex. so :

Fe^{2+}, $[Ar]3d^6$



$[FeF_6]^{4-}$



SP^3d^2 hybrid orbitals

Six pairs of electrons from six f^- IONS.

85. (2)

Sol. Properties of ionic solid

86. (2)

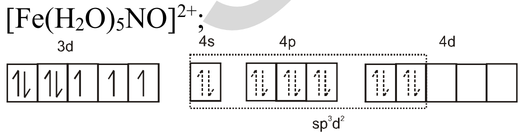
Sol. (I) Au in +3 oxidation state with $5d^8$ configuration has higher CFSE. So complex has dsp^2 hybridisation and is diamagnetic.

(II) Cu is in +1 oxidation state with $3d^{10}$ configuration and no (n –1)d orbital is available for dsp^2 hybridisaiton, so ns and np orbitals undergo sp^3 hybridisation and complex is diamagnetic.

(III) Co is in +3 oxidation state and $3d^6$ configuration has higher CFSE. So complex is diamagnetic and has d^2sp^3 hybridisation.

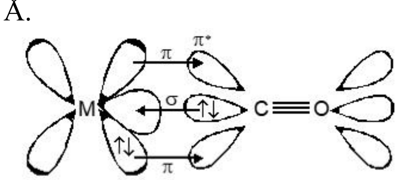
(IV) Fe is in +1 oxidation state and the complex is paramagnetic with three unpaired electrons.

$[Fe(H_2O)_5NO]^{2+}$;



87. (1)

Sol. On account of synergic interaction between metal and CO bond order of CO reduces to approximately two and half from three in carbonmonoxide. Thus bond length increases to 1.158 Å.

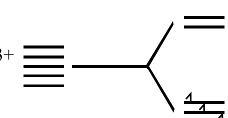


88. (2)

Sol.

(P)	$[Cr(NH_3)_4Cl_2]Cl$	Cr^{+3} is d^3. It is paramagnetic and it shows cis-trans isomerism.
(Q)	$[Ti(H_2O)_5Cl](NO_3)_2$	Ti^{+3} is d^1. It is paramagnetic and it show ionisation isomerism.
(R)	$[Pt(en)(NH_3)Cl]NO_3$	Pt^{+2} is d^8. But this complex is square planar and all electron are paired. So it is diamagnatic. It exhibit ionisation isomerism.
(S)	$[Co(NH_3)_4(NO_3)_2]NO_3$	Co^{+3} is d^6. Since ligands are strong, so electron are paired. It is diamagnetic. It exhibit cis-trans isomerism.

89. (3)

Sol. S_1 : Cr^{3+}  CFSE = 3 × – 0.4 = – 1.2 Δ_0, hybridisation is d^2sp^3 (NH_3 is strong field ligand)

S_2 : Fe^{3+}, $3d^5$ - one unpaired electron after pairing (CN^- is stronger field ligand)

$\therefore \quad \mu = \sqrt{1\ (1+2)} \approx 1.73$ BM

S_3 : $[\overset{II}{Fe}(\overset{-5}{CN})_5\overset{+}{NO}]^{2-}$ and $[\overset{II}{Fe}(\overset{-5}{CN})_5NOS^{-1}]^{4-}$.

In reactant and product, the iron is in same oxidation state i.e. +2.

90. (2)

Sol. Greater the negative charge on metal; greater is the metal-carbon bond order and lesser will be carbon-oxygen bond order.